

Independent Axial Seamount Thermomechanical Model: Progress Report

Axial Modeling Project

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Abstract

This report records an independent PyLith implementation of the written thermomechanical method for Axial Seamount, with observational checking against authorized Ocean Observatories Initiative (OOI) records and original raw BPR channels from earlier Axial deployments. It uses no paper-produced observations, corrections, numerical results, model output, or figure data. The project verifies elastic and one-branch Maxwell responses, steady three-dimensional heat conduction, temperature-property transfer, and postprocessed failure connectivity. Central and Eastern OOI histories provide 3,927 common finite daily records from 2014 through 2026; the aggregate quality code is retained as NOT EVALUATED. Raw NCEI and MGDS channels add event checks for 1998 and 2011 and deployment comparisons through 2013. The current Maxwell diagnostic fits pressure through static elastic compliance, then produces a large Central displacement error and an early modeled failure path. These outputs remain provisional: the ellipsoid compliance is not mesh-converged, generalized Maxwell branch parameters are unavailable, Eq. 16 is inconsistent with its stated modulus limits, and the paper’s written thermal equation specifies zero heat production without mechanical feedback. The current work therefore supports a partial implementation record, not a reproduction of the published eruption forecasts.

1 Scope and data provenance

The project reconstructs model inputs from written descriptions of the study and its supplement, then implements independent solvers and checks. The main model article and publisher-served supplement are specification sources [1, 2]. No observation series, corrections, numerical results, model output, source code, plotting scripts, or figure data produced for the paper are used. The report does not digitize plotted curves or use published numerical results. The supplement copy carries a confidential manuscript footer, so extracted parameters remain version-qualified.

OOI daily BPR depth records cover Central and Eastern Caldera. The products report relative uplift after detiding and drift correction; their aggregate quality flag is NOT EVALUATED [4]. The analysis retains flag code 2 without filtering. Common finite observations span 2014-09-05 through 2026-09-30. Raw NCEI records provide uncorrected pressure from the 1997–99 and 2000–02 Axial deployments. Selected MGDS files provide original `Depth` or `RawDep` channels from Center and South deployments between 2003 and 2013 [5, 6]. The workflow excludes derived detided, filtered, and drift-corrected fields; source files and processed daily series remain outside version control. The MGDS archive uses the CC BY-NC-SA 3.0 license, and its data citation and license are preserved in the repository data notes.

2 Written model and implementation

The specified reservoir is an ellipsoid measuring 6 by 3 by 1 km, centered 1.6 km below the seafloor. The written thermal method is steady conduction,

$$\nabla \cdot (k \nabla T) = -Q, \quad Q = 0, \quad (1)$$

with a 30 °C/km background geotherm, a 0 °C surface, and a 1200 °C reservoir boundary. The hydrothermal conductivity law is implemented as a separate case. Because lateral and basal thermal conditions are not specified, the current finite-element calculation extends the background geotherm to all outer faces and records this as an assumption.

The Arrhenius viscosity law is evaluated from the temperature field. The printed temperature-dependent Young’s modulus equation increases from about 25 GPa toward 75 GPa as temperature rises, contrary to the written brittle and ductile descriptions. The implementation preserves this equation only as a diagnostic; it does not claim the trend is physically resolved. The written Maxwell model requires a generalized branch spectrum that is not fully specified. Current PyLith runs therefore use one Maxwell branch with explicitly assumed properties. The PyLith 5.0.2 material interface supplies properties and initial state through an auxiliary spatial database; a same-mesh restart can transfer state when properties change [3].

The hydrothermal thermal solve now produces an exploratory midplane map of temperature-dependent properties (Fig. 1). This slice uses the written Arrhenius viscosity and conductivity relations and evaluates Eq. 16 exactly as printed. The finite-thickness slice, outer-face geotherm, and incomplete generalized Maxwell spectrum limit comparison with the complete four-rheology panels.

The project has verified a three-dimensional steady heat solve, a synthetic thermal-to-property-to-mechanics transfer, and same-mesh Maxwell restart. An unchanged-property segmented run differs from a continuous run by at most 1.612×10^{-8} in displacement, stress, total strain, and viscous strain. A separate uniform 1200 °C material swap carries viscous strain across the restart boundary and changes the final displacement by 50.08 percent relative to the uniform-property run. This verifies the restart interface; the artificial uniform-temperature swap is not a coupled physical case.

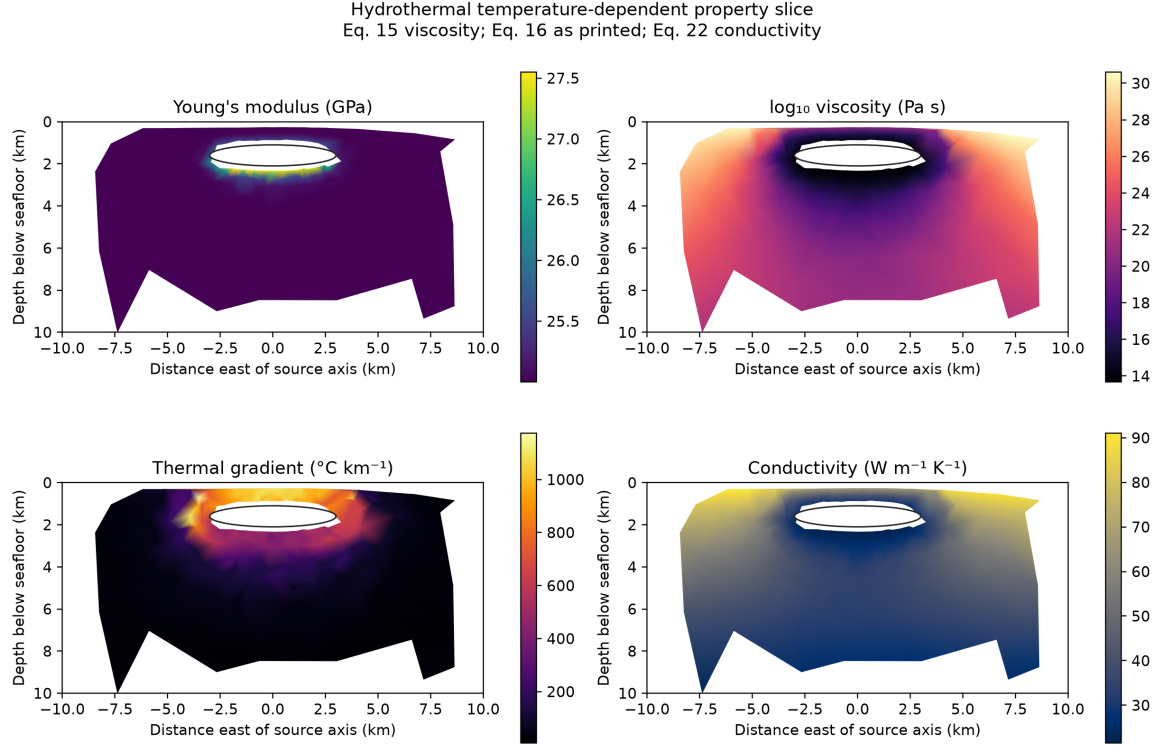
The paper’s thermal method specifies $Q = 0$ and gives no stress- or deformation-generated heat term. The roadmap’s mechanics-to-thermal return coupling therefore lacks a written physical law. The current one-way OOI diagnostic does not add an invented heat source. This prevents a complete two-way reproduction until the return coupling is specified or approved as a separate exploratory extension.

3 Independent observations and model checks

Figure 2 shows the available Central and Eastern OOI uplift records. Each instrument uses its own first-sample reference. The plot is an observation-only partial record: it has no pre-2014 series, eruption markers, or earthquake counts.

An elastic Mogi point source is calibrated to Central uplift and checked against the Eastern record. The inferred pressure spans -2.96 to $+1.06$ GPa; Eastern RMSE is 0.0594 m with 0.995 correlation. The spatial correlation does not validate this pressure scale. It shows that the point-source proxy cannot supply the intended historical ellipsoid pressure model.

The raw historical channels add Center-to-South checks at the 1998 and 2011 eruptions and three multi-year deployment overlaps. The latter span 614 paired days in 2003–05, 572 in 2007–09, and 731 in 2011–13. Static Mogi predictions have South RMSE values of 0.134, 0.156, and 0.387 m, with respective biases of +0.113, +0.136, and +0.363 m. These channels retain tides, oceanographic variability, and



Cell-center projection within $|y| \leq 0.7$ km; white ellipse marks the reservoir. Side and basal temperatures use the assumed 30°C/km geotherm. Eq. 16 conflicts with the stated brittle and ductile limits.

Figure 1: Hydrothermal temperature-dependent properties on a finite-thickness slice around the model $y=0$ plane. Cell-center values show Young's modulus from Eq. 16 as printed, Arrhenius viscosity, the finite-element temperature gradient, and Eq. 22 conductivity. The white ellipse marks the reservoir. Side and basal temperatures extend the assumed 30°C/km geotherm. Generated by `make thermal-property-slices`; Eq. 16 remains an unresolved transcription diagnostic.

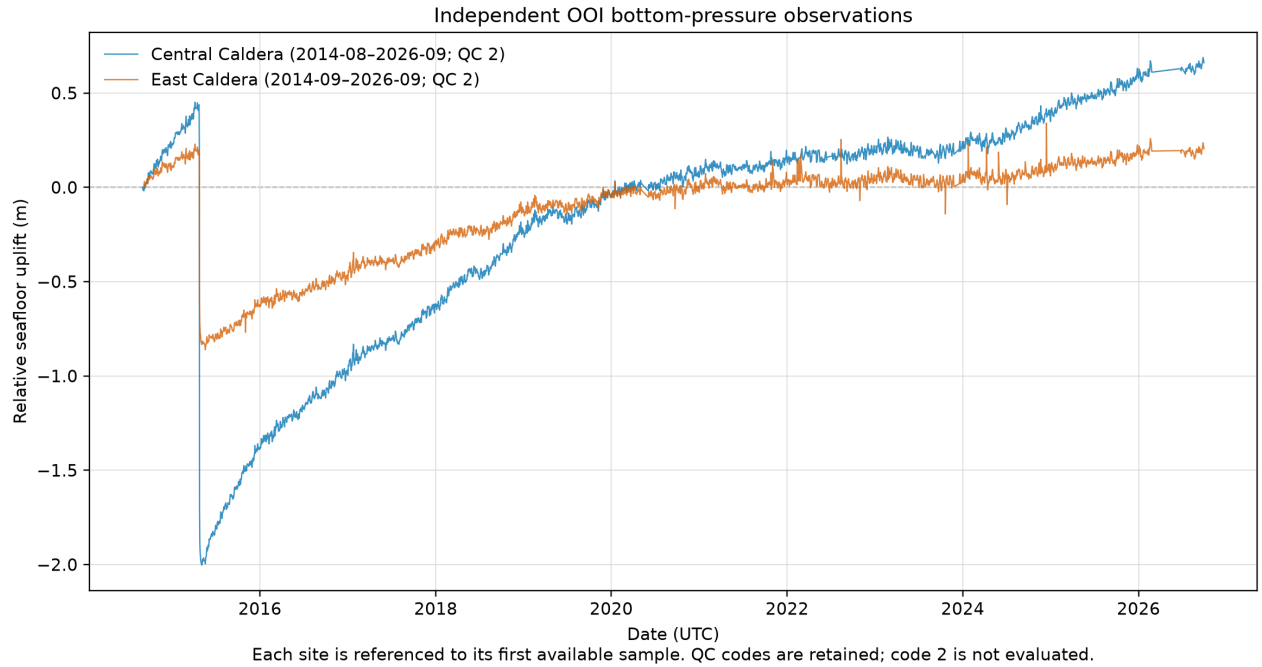


Figure 2: Daily OOI relative uplift at Central and Eastern Caldera from each site’s first available record. Generated by `make bpr-observation-plot`; aggregate quality code 2 (NOT EVALUATED) is retained. The data show the observed interval available for independent model checking.

sensor drift, and the fits omit viscoelastic memory. They extend observation checking across the pre-OOI interval; they are not calibrated pressure histories or eruption forecasts.

The elastic ellipsoid comparison improves the pressure scale but remains mesh-sensitive. On the 2,761-tetrahedron mesh, unit-load compliance is 0.0318994 m/MPa at Central and 0.00345580 m/MPa at Eastern. The static Central-calibrated pressure spans -62.21 to 22.16 MPa. Eastern RMSE is 0.221 m, relative L2 error is 76.36 percent, and correlation is 0.995. Across tested refinements, station compliance changes by 21–76 percent; these results do not establish a stable pressure calibration.

The one-branch Maxwell diagnostic uses monthly pressure inferred from the static elastic Central compliance, linearly interpolated between samples. Its 147 stress records cover 12.07 years. Central RMSE is 1.096 m (correlation 0.787); Eastern RMSE is 0.195 m (correlation 0.922). The Mohr–Coulomb post-processor uses $C = 1$ MPa, applies the tabulated 25 degrees directly as the friction angle, assumes zero pore pressure, and applies no tensile cutoff because tensile strength is unavailable. It finds a cavity-to-surface path in 146 records, first at a saved time of 60 days. These are diagnostic outcomes, not eruption predictions.

4 Numerical verification

The bounded PyLith 5.0.2 toolchain runs an elastic-cavity smoke case on 2,761 linear tetrahedra. Analytical Mogi checks verify positive inflation uplift, radial symmetry, and pressure scaling. A PyLith spherical-cavity comparison has a 33.2 percent nearest-axis error on 3,191 tetrahedra; this coarse check does not establish mesh convergence. The tetrahedral thermal solver matches manufactured linear and uniform-source solutions. Its full ellipsoid thermal case converges numerically, but its lateral and basal boundary temperatures remain assumptions.

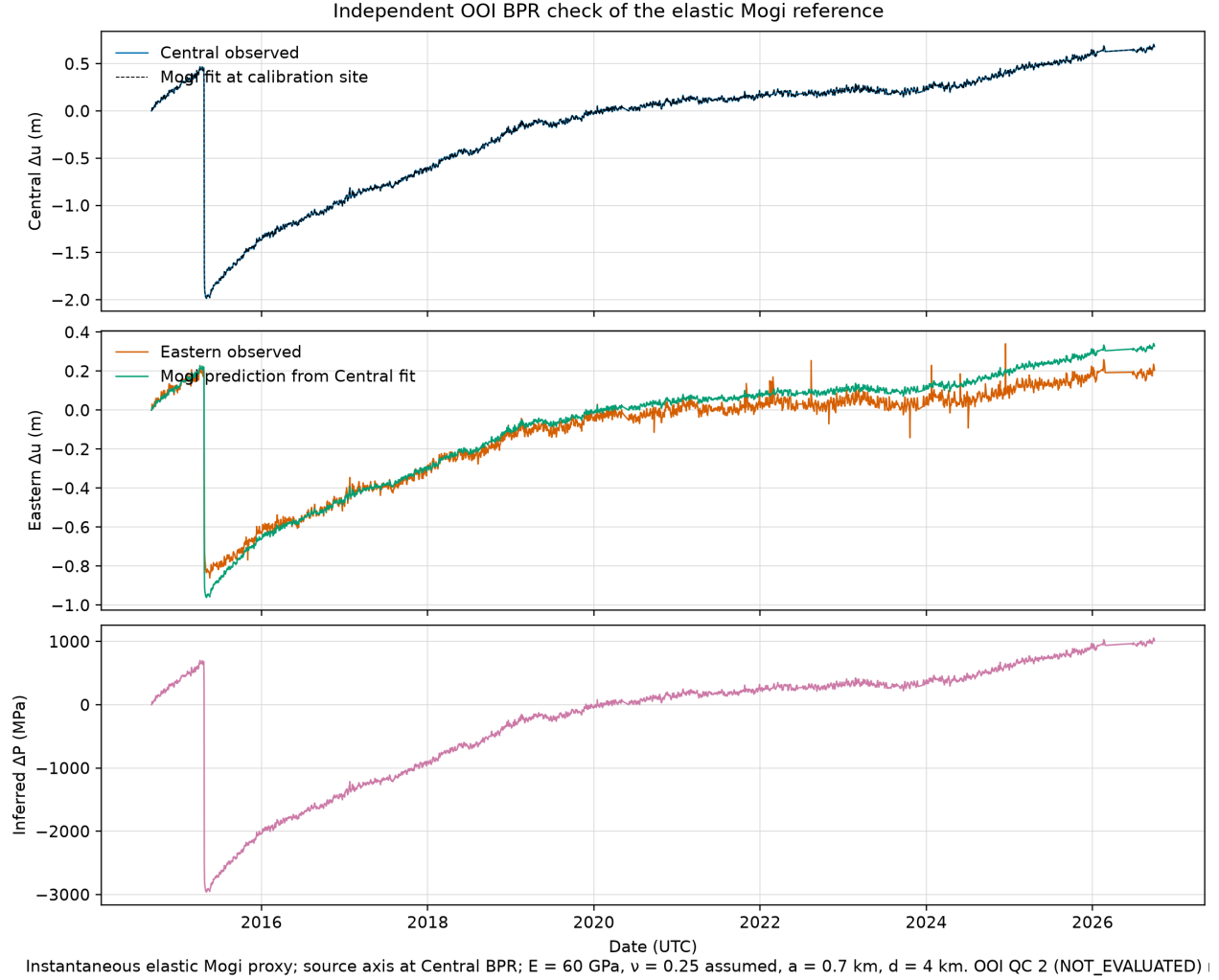


Figure 3: OOI observations and elastic Mogi point-source response. Generated by `make bpr-mogi-check` from independent OOI records; Central is fit by construction and Eastern is held out. The inferred pressure scale is physically unsuitable for the later reservoir model.

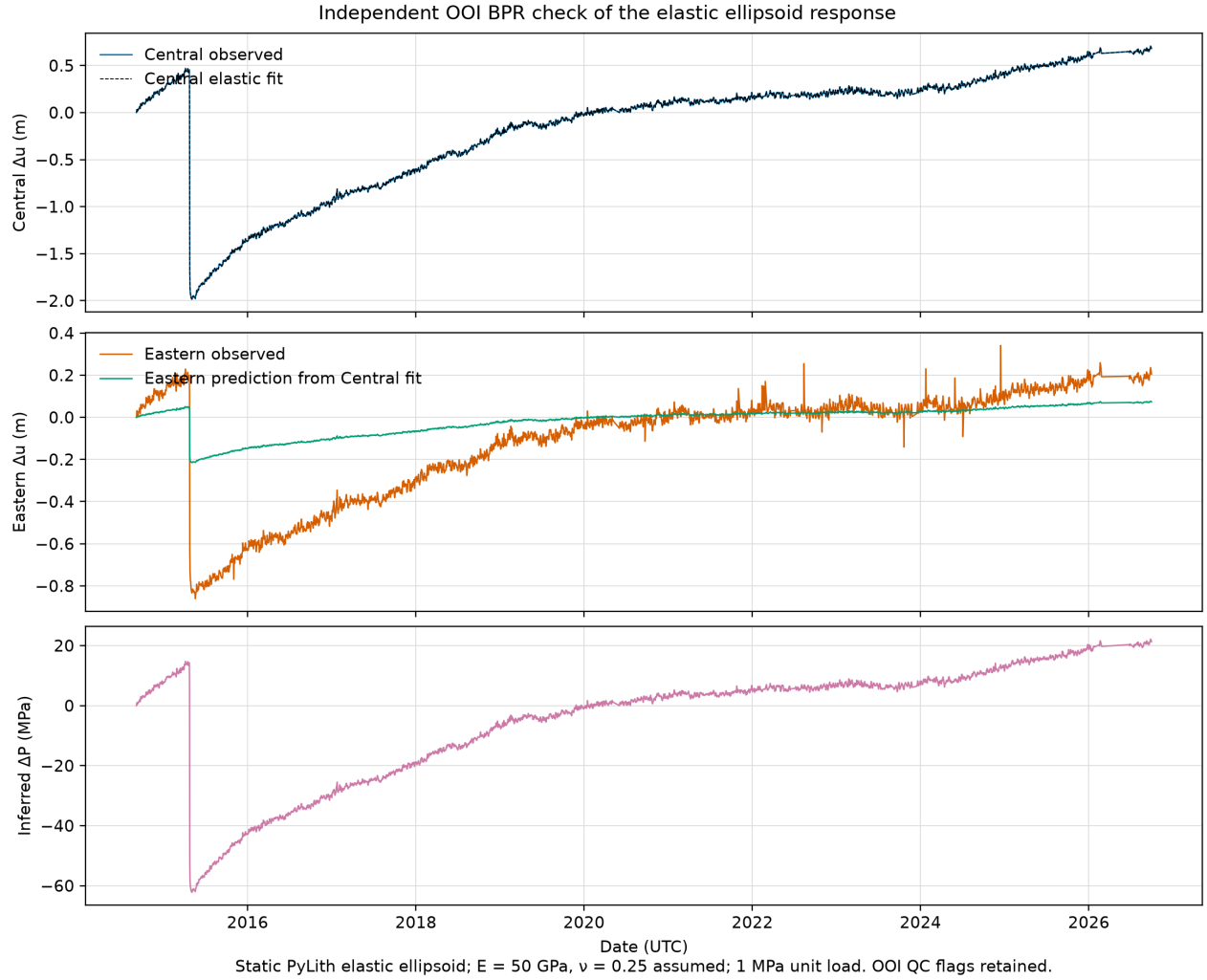
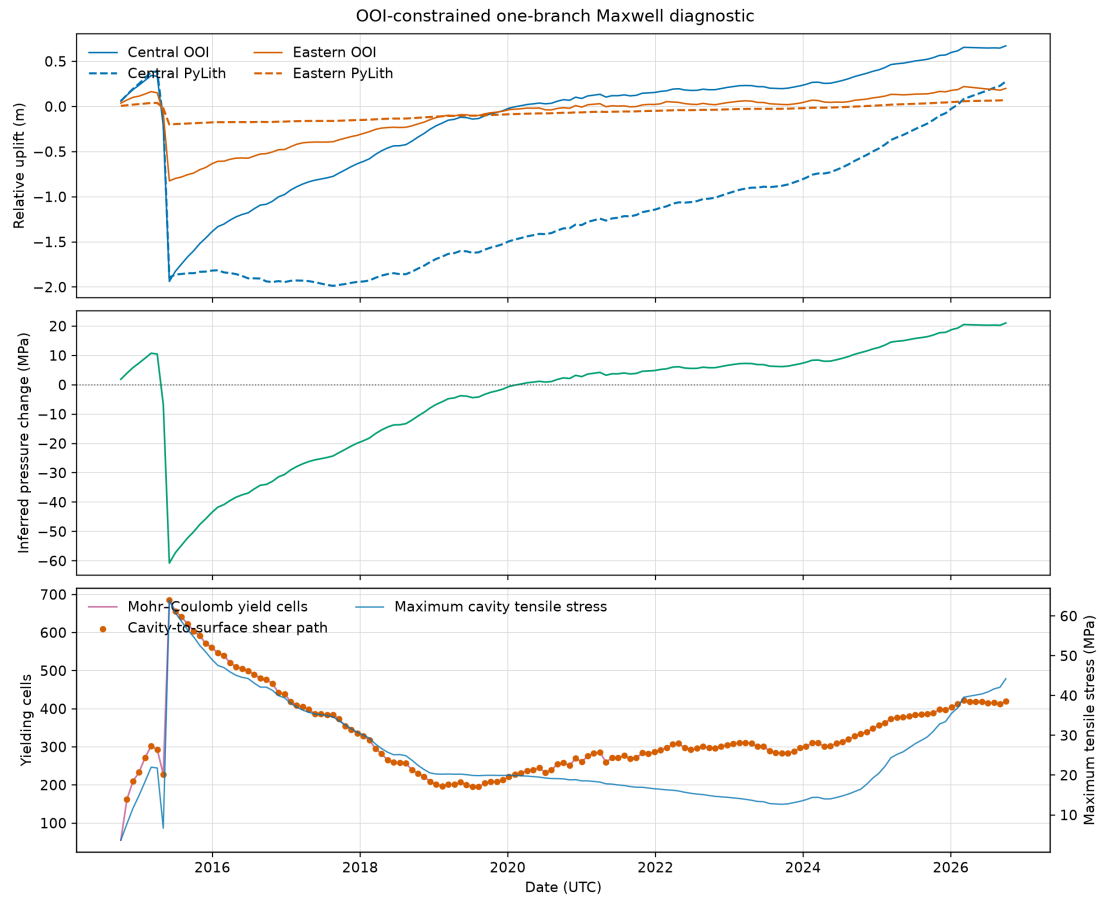


Figure 4: Static elastic ellipsoid fit to Central OOI uplift and Eastern holdout prediction. Generated by `make ellipsoid-bpr-check`. The amplitude mismatch and nonconverged compliance make the inferred pressure provisional.



OOI QC 2 (NOT_EVALUATED) retained. Pressure uses static elastic Central compliance. Failure uses $C = 1$ MPa, 25° directly, zero pore pressure, and no tensile cutoff. The 2,761-tetrahedron compliance is not mesh-converged.

Figure 5: OOI-constrained one-branch Maxwell diagnostic. The panels compare OOI and modeled uplift, static-fit pressure, and stress-derived failure indicators. Generated by `make ooi-maxwell-history-plot`. Quality code 2 is retained; pressure uses static compliance; the mesh and failure assumptions remain provisional.

The failure analysis converts PyLith Cauchy stress to principal stresses, applies a Mohr–Coulomb yield test, and searches face-connected tetrahedra between the cavity and surface. Synthetic tests verify state ordering, boundary identification, and time-resolved path detection. Those checks validate software behavior under stated assumptions; they do not validate a physical failure threshold.

5 Panel-by-panel status

No manuscript numerical panel is independently reproduced. Figure 2 is an observation-only partial record; Figures 3–5 are project-generated diagnostics. The detailed command, output, and comparison inventory is maintained in `docs/figure_reproduction.md`.

Panel	Status and current limitation
Fig. 1a	Not reproduced; required bathymetry, lava-flow, and event source records are excluded.
Fig. 1b	Not reproduced; a schematic has not been generated and the complete boundary setup is unresolved.
Fig. 2	Partially reproduced from OOI observations for 2014–2026; eruption markers and event counts are absent.
Fig. 3, elastic	Not reproduced; the required property slices and full written configuration are unavailable.
Fig. 3, viscoelastic	Not reproduced; a one-branch smoke test does not provide generalized rheology or panel fields.
Fig. 3, temperature-dependent	Not reproduced; Eq. 16 is inconsistent with its stated limits and the fields are incomplete.
Fig. 3, hydrothermal	Partially reproduced; one hydrothermal property slice is available, but the full rheology panels and resolved Eq. 16 are not.
Fig. 4a	Not reproduced; OOI diagnostics cover only 2014–2026 and the static compliance is not mesh-converged.
Fig. 4b	Not reproduced; the full calibrated reservoir-volume and historical deformation record is unavailable.
Fig. 5a	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5b	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5c	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5d	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5e	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5f	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5g	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5h	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5i	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5j	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5k	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5l	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5m	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5n	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5o	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5p	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5q	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5r	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5s	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 5t	Not reproduced; event-scale stress history and rheology mapping are unavailable.
Fig. 6a–d	Not reproduced; no conceptual redraw has been made.
Supp. S1	Partially reproduced for one hydrothermal temperature-dependent case; complete slices and all four rheologies are not available.

Panel	Status and current limitation
Supp. S2	Not reproduced; the hydrothermal brittle–ductile transition comparison is incomplete.
Supp. S3	Not reproduced; independent Del Negro and prior FEM benchmarks are incomplete.
Supp. S4	Not reproduced; a verified Winkler-foundation implementation is unavailable.
Supp. S5	Not reproduced; the complete boundary and tectonic-loading convention is unresolved.
Supp. S6	Not reproduced; mesh-converged OOI pressure calibration and the full rheology set are unavailable.

6 Limitations and next work

The present implementation is a collection of verified components, OOI checks, and raw historical BPR diagnostics. It does not implement the four rheology configurations, a generalized Maxwell spectrum, a verified Winkler foundation, or the complete historical pressure calibration. The model-box dimensions, Poisson ratio, host density, tensile strength, and several source conventions remain unresolved. The finite-element compliance is not mesh-converged.

The written thermal equation is steady and sets heat production to zero, while the project roadmap asks for mechanics-to-thermal feedback. No return-coupling law is specified in the written model. This report preserves that gap instead of silently inventing a source term. A scientifically faithful reproduction cannot claim full two-way coupling until the governing feedback and missing parameters are specified. Raw historical BPR checks add eruption and inter-eruption observations, but their uncorrected channels and static elasticity do not reproduce the 1998–2011 stress evolution.

References

1. Cabaniss, H. E., Gregg, P. M., Nooner, S. L., and Chadwick, W. W. (2020). *Triggering of eruptions at Axial Seamount, Juan de Fuca Ridge*. Scientific Reports, 10, 10219. doi:[10.1038/s41598-020-67043-0](https://doi.org/10.1038/s41598-020-67043-0). Used as a written method specification; publication data and figures were not used.
2. Publisher-served supplementary information for Cabaniss et al. (2020). <https://doi.org/10.1038/s41598-020-67043-0>. The served PDF carries a confidential-manuscript footer; extracted values may represent a pre-publication version.
3. PyLith 5.0.2 User Guide, <https://pylith.readthedocs.io/en/v5.0.2/>.
4. Ocean Observatories Initiative, BOTSFLU daily depth product, <https://oceanobservatories.org/data-product/botsflu/>.
5. National Oceanic and Atmospheric Administration, National Centers for Environmental Information (2005), Deep-Ocean Assessment and Reporting of Tsunamis (DART) raw bottom-pressure data, doi:<https://doi.org/10.7289/V5F18WNS>.
6. Chadwick, W., et al. (2023), Processed Bottom Pressure Recorder data from uncabled instruments deployed at Axial Seamount, Marine Geoscience Data System, doi:<https://doi.org/10.1594/IEDA/322282>. The analysis reads only original raw `Depth` and `RawDep` channels and excludes the accompanying corrections.